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Section:6

OSSP (Operating Systems And Systems Programming) - 25CS2104E

(SKILL Content Delivery Concepts)

3.To Apply Escape Sequences, Store Command History, Navigate Previous Commands, Navigate Next Commands, Update Input Buffer, Test Recall Functionality.

To Allocate Buffers Dynamically, Resize Arrays, Prevent Buffer Overflow, Manage Linked Lists, Release Memory Correctly, Verify with Valgrind.

Step 1- Create the folder

```
monis@DESKTOP-C3GM4T6: ~/OSLAB/skill
monis@DESKTOP-C3GM4T6:~$ ls
Linux-System-Programming-Week1  OSSP-Practical      foldername  moni    sample1.c
OSLAB                          Simple-Commands-Week1 hello       monis   sample1.c.save
OSLAB14                        example.txt         hello.c     sample1
monis@DESKTOP-C3GM4T6:~$ cd OSLAB
monis@DESKTOP-C3GM4T6:~/OSLAB$ ls
OSLAB28  Scheduling-Algorithms  hello  hello.c:Zone.Identifier  hello_link  newfile.txt  test1
Pipes    file1.txt              hello.c  hello:Zone.Identifier    hello_shortcut  skill
monis@DESKTOP-C3GM4T6:~/OSLAB$ cd skill
```

Step 2 — Create the C file

```
monis@DESKTOP-C3GM4T6: ~/OSLAB/skill
monis@DESKTOP-C3GM4T6:~$ ls
Linux-System-Programming-Week1  OSSP-Practical      foldername  moni    sample1.c
OSLAB                          Simple-Commands-Week1 hello       monis   sample1.c.save
OSLAB14                        example.txt         hello.c     sample1
monis@DESKTOP-C3GM4T6:~$ cd OSLAB
monis@DESKTOP-C3GM4T6:~/OSLAB$ ls
OSLAB28  Scheduling-Algorithms  hello  hello.c:Zone.Identifier  hello_link  newfile.txt  test1
Pipes    file1.txt              hello.c  hello:Zone.Identifier    hello_shortcut  skill
monis@DESKTOP-C3GM4T6:~/OSLAB$ cd skill
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$ nano history.c
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$
```

```
monis@DESKTOP-C3GM4T6: ~/OSLAB/skill
GNU nano 8.7.1 history.c *
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <termios.h>
#include <unistd.h>

#define MAX_HISTORY 20
#define MAX_LENGTH 100

char history[MAX_HISTORY][MAX_LENGTH];
int history_count = 0;

void disable_raw_mode(struct termios *old)
{
    tcsetattr(STDIN_FILENO, TCSAFLUSH, old);
}

void enable_raw_mode(struct termios *old)
{
    struct termios raw;

    tcgetattr(STDIN_FILENO, old);

    raw = *old;
    raw.c_lflag &= ~(ICANON | ECHO);
    raw.c_cc[VMIN] = 1;
    raw.c_cc[VTIME] = 0;
    tcsetattr(STDIN_FILENO, TCSAFLUSH, &raw);
}

void show_command(const char *command)
{
    printf("\r\033[Kmyshe11> %s", command);
    fflush(stdout);
}

int main()
{
    char buffer[MAX_LENGTH];
    int length;
    int history_index;
    char ch;

    struct termios old;

^G Help      ^O Write Out ^F Where Is  ^K Cut       ^T Execute   ^C Location  M-U Undo
^X Exit      ^R Read File ^\ Replace   ^U Paste     ^J Justify   ^_ Go To Line M-E Redo
```

```
monis@DESKTOP-C3GM4T6: ~/OSLAB/skill
GNU nano 8.7.1 history.c *
struct termios old;

while (1)
{
    memset(buffer, 0, sizeof(buffer));

    length = 0;

    history_index = history_count;

    printf("myshell> ");
    fflush(stdout);

    enable_raw_mode(&old);

    while (1)
    {
        read(STDIN_FILENO, &ch, 1);

        /* Enter key */
        if (ch == '\n' || ch == '\r')
        {
            buffer[length] = '\0';

            printf("\n");

            break;
        }

        /* Backspace */
        if (ch == 127 || ch == '\b')
        {
            if (length > 0)
            {
                length--;

                buffer[length] = '\0';

                printf("\b \b");

                fflush(stdout);
            }

            continue;
        }
    }
}

^G Help      ^O Write Out  ^F Where Is   ^K Cut        ^T Execute    ^C Location   M-U Undo
^X Exit      ^R Read File  ^\ Replace    ^U Paste      ^J Justify    ^/_ Go To Line M-E Redo
```

```
monis@DESKTOP-C3GM4T6: ~/OSLAB/skill
GNU nano 8.7.1 history.c
}

    continue;
}

/* Escape sequence */
if(ch == 27)
{
    char seq[2];
    if(read(STDIN_FILENO,& seq[0],1)== 1 &&
        read(STDIN_FILENO, &seq[1],1)== 1)
    {
        /* Up Arrow */
        if (seq[0] == '[' && seq[1] == 'A')
        {
            if (history_index > 0)
            {
                history_index--;

                strcpy(buffer, history[history_index]);

                length = strlen(buffer);

                show_command(buffer);
            }
        }

        /* Down Arrow */
        else if (seq[0] == '[' && seq[1] == 'B')
        {
            if (history_index < history_count - 1)
            {
                history_index++;

                strcpy(buffer, history[history_index]);

                length = strlen(buffer);

                show_command(buffer);
            }
        }
        else
        {
            history_index = history_count;

            buffer[0] = '\0';
        }
    }
}

^G Help      ^O Write Out  ^F Where Is   ^K Cut        ^T Execute    ^C Location   M-U Undo
^X Exit      ^R Read File  ^\ Replace    ^U Paste      ^J Justify    ^_ Go To Line  M-E Redo
```

```
        buffer[0] = '\0';

        length = 0;

        show_command(buffer);
    }
}

continue;
}

/* Store normal characters */
if (length < MAX_LENGTH - 1)
{
    buffer[length++] = ch;

    buffer[length] = '\0';

    putchar(ch);

    fflush(stdout);
}

disable_raw_mode(&old);

/* Exit condition */
if (strcmp(buffer, "exit") == 0)
{
    printf("Exiting shell...\n");

    break;
}

/* Ignore empty command */
if (length == 0)
{
    continue;
}

/* Store command in history */
if (history_count < MAX_HISTORY)
{
    strcpy(history[history_count], buffer);
```

^G Help
^X Exit

^O Write Out
^R Read File

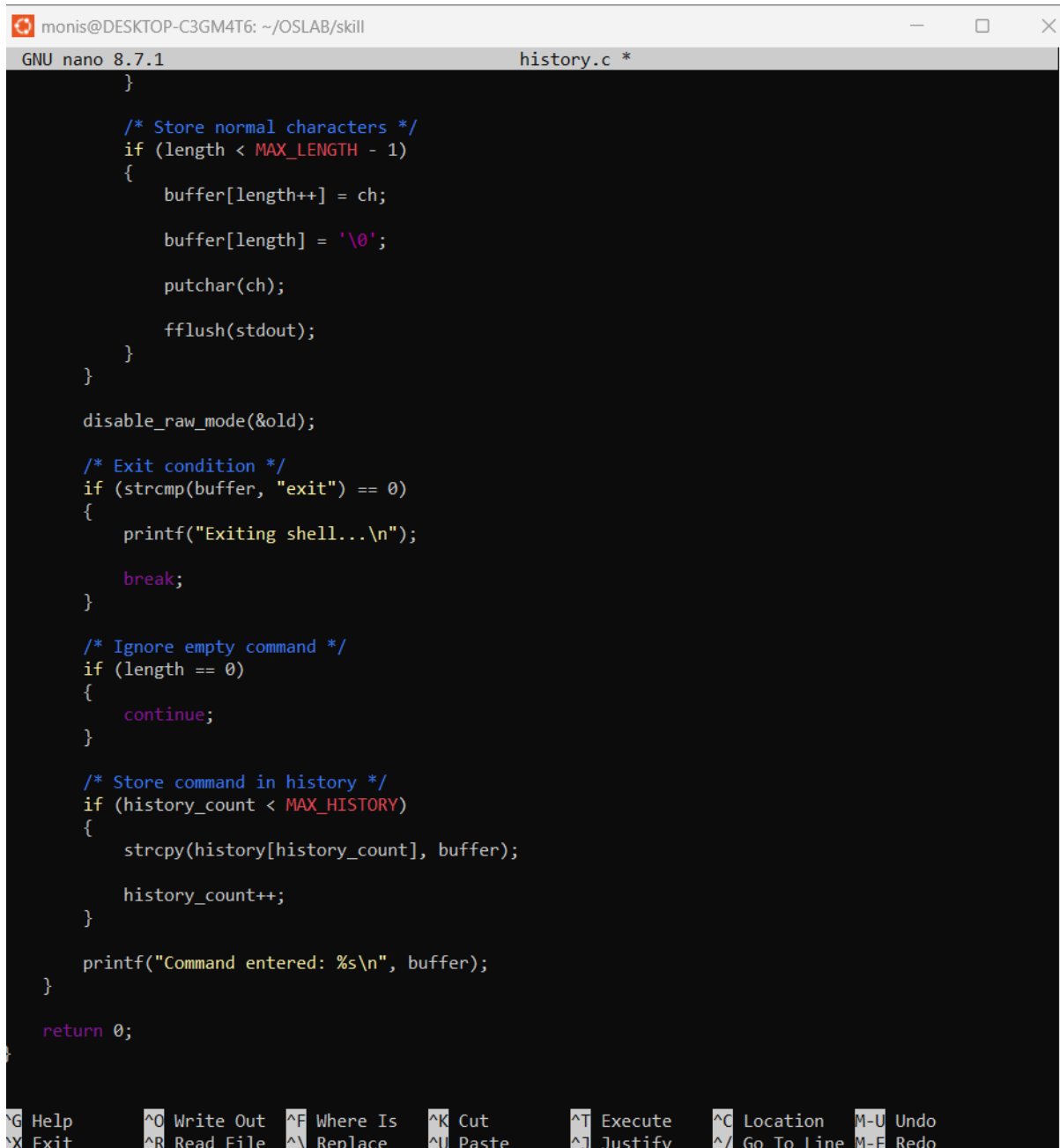
^F Where Is
^_ Replace

^K Cut
^U Paste

^T Execute
^J Justify

^C Location
^_ Go To Line

M-U Undo
M-E Redo



```
monis@DESKTOP-C3GM4T6: ~/OSLAB/skill
GNU nano 8.7.1 history.c *
}

/* Store normal characters */
if (length < MAX_LENGTH - 1)
{
    buffer[length++] = ch;

    buffer[length] = '\0';

    putchar(ch);

    fflush(stdout);
}

disable_raw_mode(&old);

/* Exit condition */
if (strcmp(buffer, "exit") == 0)
{
    printf("Exiting shell...\n");

    break;
}

/* Ignore empty command */
if (length == 0)
{
    continue;
}

/* Store command in history */
if (history_count < MAX_HISTORY)
{
    strcpy(history[history_count], buffer);

    history_count++;
}

printf("Command entered: %s\n", buffer);
}

return 0;
}

^G Help      ^O Write Out  ^F Where Is  ^K Cut       ^T Execute   ^C Location  M-U Undo
^X Exit      ^R Read File  ^\ Replace   ^U Paste     ^J Justify   ^_ Go To Line M-E Redo
```

Step 3 — Save the program

In nano:

Press:

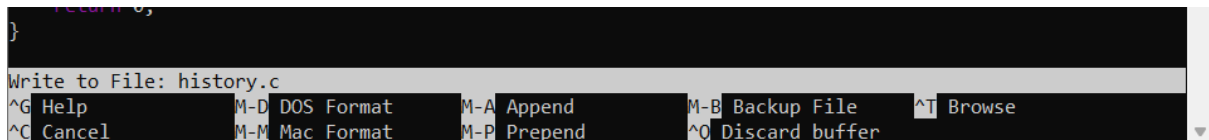
Ctrl + O

Then press:

Enter

Then:

Ctrl + X



Now check:

ls

You should see:

history.c

```
monis@DESKTOP-C3GM4T6: ~/OSLAB/skill
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$ ls
history.c  process  process.c  shell  shell.c
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$ ls -l
total 48
-rw-r--r-- 1 monis monis 3700 Sep 14 10:31 history.c
-rwxr-xr-x 1 monis monis 16392 Aug 24 16:05 process
-rw-r--r-- 1 monis monis 894 Aug 24 16:03 process.c
-rwxr-xr-x 1 monis monis 16352 Aug 24 16:50 shell
-rw-r--r-- 1 monis monis 1789 Aug 24 16:49 shell.c
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$
```

Step 4 — Compile

Run:

gcc history.c -o history

If there is no error, check:

ls -l

You should see:

history.c

history

Here:

- history.c = source code
- history = executable program

```
monis@DESKTOP-C3GM4T6: ~/OSLAB/skill
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$ gcc history.c -o history
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$ ls -l
total 68
-rwxr-xr-x 1 monis monis 16672 Sep 14 10:37 history
-rw-r--r-- 1 monis monis 3700 Sep 14 10:31 history.c
-rwxr-xr-x 1 monis monis 16392 Aug 24 16:05 process
-rw-r--r-- 1 monis monis 894 Aug 24 16:03 process.c
-rwxr-xr-x 1 monis monis 16352 Aug 24 16:50 shell
-rw-r--r-- 1 monis monis 1789 Aug 24 16:49 shell.c
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$
```

Step 5 — Run the program

Run:

`./history`

You should get:

`myshell>`

Now type:

`pwd`

Press Enter.

You should see something like:

`myshell> pwd`

Command entered: pwd

`myshell>`

```
monis@DESKTOP-C3GM4T6: ~/OSLAB/skill
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$ ./history
myshell> pwd
Command entered: pwd
myshell>
```

Step 6 — Test command history

Now enter three commands.

First:

`pwd`

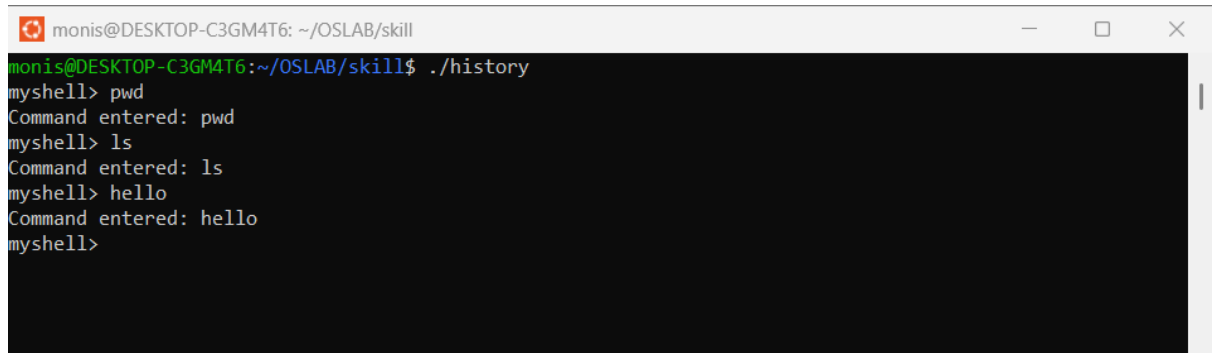
Second:

ls

Third:

hello

These commands are now stored in the history array.

A terminal window titled 'monis@DESKTOP-C3GM4T6: ~/OSLAB/skill'. The prompt is 'monis@DESKTOP-C3GM4T6:~/OSLAB/skill\$'. The user has entered './history', which displays a list of commands: 'myshell> pwd', 'Command entered: pwd', 'myshell> ls', 'Command entered: ls', 'myshell> hello', 'Command entered: hello', and 'myshell>'.

```
monis@DESKTOP-C3GM4T6: ~/OSLAB/skill
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$ ./history
myshell> pwd
Command entered: pwd
myshell> ls
Command entered: ls
myshell> hello
Command entered: hello
myshell>
```

Step 7 — Test UP Arrow

At:

myshell>

press:

↑

You should get:

myshell> hello

Press ↑ again:

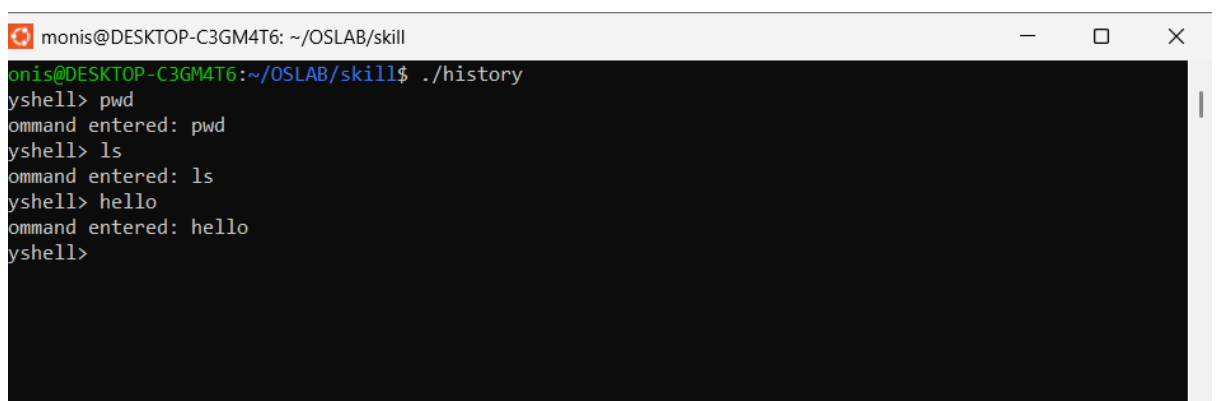
myshell> ls

Press ↑ again:

myshell> pwd

This demonstrates:

Navigate Previous Commands

A terminal window titled 'monis@DESKTOP-C3GM4T6: ~/OSLAB/skill'. The prompt is 'monis@DESKTOP-C3GM4T6:~/OSLAB/skill\$'. The user has entered './history', which displays a list of commands: 'yshell> pwd', 'ommand entered: pwd', 'yshell> ls', 'ommand entered: ls', 'yshell> hello', 'ommand entered: hello', and 'yshell>'.

```
monis@DESKTOP-C3GM4T6: ~/OSLAB/skill
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$ ./history
yshell> pwd
ommand entered: pwd
yshell> ls
ommand entered: ls
yshell> hello
ommand entered: hello
yshell>
```

```
monis@DESKTOP-C3GM4T6: ~/OSLAB/skill
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$ ./history
myshell> pwd
Command entered: pwd
myshell> ls
Command entered: ls
myshell> hello
Command entered: hello
myshell> hello
```

```
monis@DESKTOP-C3GM4T6: ~/OSLAB/skill
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$ ./history
myshell> pwd
Command entered: pwd
myshell> ls
Command entered: ls
myshell> hello
Command entered: hello
myshell> ls
```

```
monis@DESKTOP-C3GM4T6: ~/OSLAB/skill
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$ ./history
myshell> pwd
Command entered: pwd
myshell> ls
Command entered: ls
myshell> hello
Command entered: hello
myshell> pwd
```

Step 8 — Test DOWN Arrow

Suppose you are currently viewing:

```
myshell> pwd
```

Press:

↓

You should move forward in history:

```
myshell> ls
```

Press:

↓

You should get:

```
myshell> hello
```

Press ↓ again.

The input should become empty:

```
myshell>
```

This demonstrates:

Navigate Next Commands

```
monis@DESKTOP-C3GM4T6: ~/OSLAB/skill$ ./history
myshell> pwd
Command entered: pwd
myshell> ls
Command entered: ls
myshell> hello
Command entered: hello
myshell> pwd
```

```
monis@DESKTOP-C3GM4T6: ~/OSLAB/skill$ ./history
myshell> pwd
Command entered: pwd
myshell> ls
Command entered: ls
myshell> hello
Command entered: hello
myshell> ls
```

```
monis@DESKTOP-C3GM4T6: ~/OSLAB/skill$ ./history
myshell> pwd
Command entered: pwd
myshell> ls
Command entered: ls
myshell> hello
Command entered: hello
myshell> hello
```

```
monis@DESKTOP-C3GM4T6: ~/OSLAB/skill$ ./history
yshell> pwd
ommand entered: pwd
yshell> ls
ommand entered: ls
yshell> hello
ommand entered: hello
yshell>
```

Step 9 — Understand the escape sequence

```
monis@DESKTOP-C3GM4T6: ~/OSLAB/skill
monis@DESKTOP-C3GM4T6:~$ ls
Linux-System-Programming-Week1  OSLAB14      Simple-Commands-Week1  foldername  hello.c  monis  sample1.c
OSLAB                          OSSP-Practical  example.txt           hello       moni    sample1  sample1.c.save
monis@DESKTOP-C3GM4T6:~$ cd OSLAB
monis@DESKTOP-C3GM4T6:~/OSLAB$ ls
OSLAB28  Scheduling-Algorithms  hello  hello.c:Zone.Identifier  hello_link  newfile.txt  test1
Pipes    file1.txt              hello.c  hello:Zone.Identifier    hello_shortcut  skill
monis@DESKTOP-C3GM4T6:~/OSLAB$ cd skill
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$ ls
history  history.c  process  process.c  shell  shell.c
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$ nano history.c
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$ gcc history.c -o history
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$ ./history
myshell> pwd
Command entered: pwd
myshell> ls
Command entered: ls
myshell> hello
Command entered: hello
myshell> pwd
```

Output: The Up Arrow key generated the escape sequence ESC [A, which was detected by the program to recall the previous command from command history. The Down Arrow key uses ESC [B to navigate to the next command.

Step 10- Update Input Buffer

```
monis@DESKTOP-C3GM4T6: ~/OSLAB/skill
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$ nano history.c
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$ gcc history.c -o history
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$ ./history
myshell> pwd
Command entered: pwd
myshell> ls
Command entered: ls
myshell>
```

Press ↑:

```
monis@DESKTOP-C3GM4T6: ~/OSLAB/skill
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$ nano history.c
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$ gcc history.c -o history
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$ ./history
myshell> pwd
Command entered: pwd
myshell> ls
Command entered: ls
myshell> ls
```

Step 11 – Test Recall Functionality

↑ → **hello**

↑ → **ls**

↑ → **pwd**

↓ → **ls**

↓ → **hello**

↓ → **empty**

```
monis@DESKTOP-C3GM4T6: ~/OSLAB/skill
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$ ./history
myshell> pwd
Command entered: pwd
myshell> ls
Command entered: ls
myshell> hello
Command entered: hello
myshell>
```

```
monis@DESKTOP-C3GM4T6: ~/OSLAB/skill
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$ ./history
myshell> pwd
Command entered: pwd
myshell> ls
Command entered: ls
myshell> hello
Command entered: hello
myshell> hello
```

```
monis@DESKTOP-C3GM4T6: ~/OSLAB/skill
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$ ./history
myshell> pwd
Command entered: pwd
myshell> ls
Command entered: ls
myshell> hello
Command entered: hello
myshell> ls
```

```
monis@DESKTOP-C3GM4T6: ~/OSLAB/skill
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$ ./history
myshell> pwd
Command entered: pwd
myshell> ls
Command entered: ls
myshell> hello
Command entered: hello
myshell> pwd
```

```
Select monis@DESKTOP-C3GM4T6: ~/OSLAB/skill
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$ ./history
myshell> pwd
Command entered: pwd
myshell> ls
Command entered: ls
myshell> hello
Command entered: hello
myshell> pwd
```

```
monis@DESKTOP-C3GM4T6: ~/OSLAB/skill
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$ ./history
myshell> pwd
Command entered: pwd
myshell> ls
Command entered: ls
myshell> hello
Command entered: hello
myshell> ls
```

```
monis@DESKTOP-C3GM4T6: ~/OSLAB/skill
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$ ./history
myshell> pwd
Command entered: pwd
myshell> ls
Command entered: ls
myshell> hello
Command entered: hello
myshell> hello
```

```
monis@DESKTOP-C3GM4T6: ~/OSLAB/skill
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$ ./history
yshell> pwd
ommand entered: pwd
yshell> ls
ommand entered: ls
yshell> hello
ommand entered: hello
yshell>
```

Step 12 — Exit the program

```
monis@DESKTOP-C3GM4T6: ~/OSLAB/skill
monis@DESKTOP-C3GM4T6:~$ ls
Linux-System-Programming-Week1  OSLAB14      Simple-Commands-Week1  foldername  hello.c  monis  sample1.c
OSLAB                          OSSP-Practical  example.txt            hello       moni    sample1  sample1.c.save
monis@DESKTOP-C3GM4T6:~$ cd OSLAB
monis@DESKTOP-C3GM4T6:~/OSLAB$ ls
OSLAB28  Scheduling-Algorithms  hello      hello.c:Zone.Identifier  hello_link  newfile.txt  test1
Pipes    file1.txt              hello.c    hello:Zone.Identifier    hello_shortcut  skill
monis@DESKTOP-C3GM4T6:~/OSLAB$ cd skill
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$ ls
history  history.c  process  process.c  shell  shell.c
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$ nano history.c
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$ gcc history.c -o history
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$ ./history
myshell> pwd
Command entered: pwd
myshell> ls
Command entered: ls
myshell> hello
Command entered: hello
myshell> pwd
Command entered: pwd
myshell> exit
```

```

monis@DESKTOP-C3GM4T6:~/OSLAB/skill$ nano history.c
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$ gcc history.c -o history
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$ ./history
myshell> pwd
Command entered: pwd
myshell> ls
Command entered: ls
myshell> hello
Command entered: hello
myshell> pwd
Command entered: pwd
myshell> exit
Exiting shell...
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$

```

Step 13: Allocate Buffers Dynamically

Use dynamic memory allocation with `malloc()`.

```

monis@DESKTOP-C3GM4T6: ~/OSLAB/skill
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$ nano memory.c
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$

```

```

monis@DESKTOP-C3GM4T6: ~/OSLAB/skill
GNU nano 8.7.1 memory.c *
#include <stdio.h>
#include <stdlib.h>

int main() {
    int size = 5;

    int *buffer = (int *)malloc(size * sizeof(int));

    if (buffer == NULL) {
        printf("Memory allocation failed.\n");
        return 1;
    }

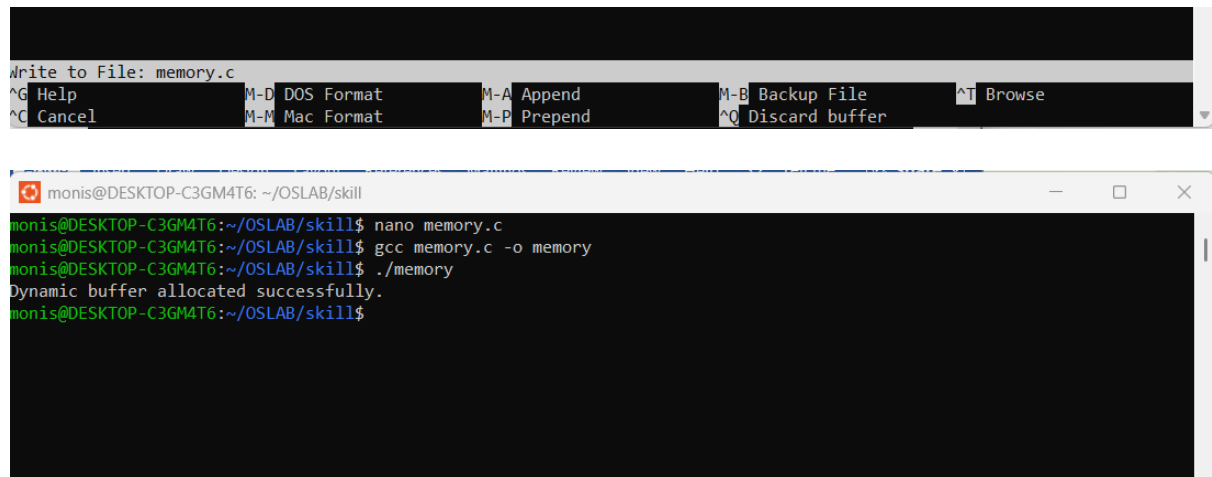
    printf("Dynamic buffer allocated successfully.\n");

    free(buffer);

    return 0;
}

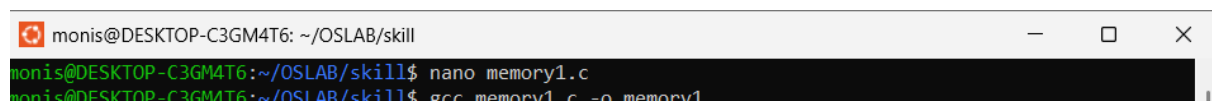
^G Help      ^O Write Out  ^F Where Is   ^K Cut        ^T Execute    ^C Location   M-U Undo      M-A
^X Exit      ^R Read File  ^\ Replace    ^U Paste      ^J Justify    ^_ Go To Line  M-E Redo      M-6

```



Step 14: Resize Arrays

To demonstrate resizing, use `realloc()`.



```
monis@DESKTOP-C3GM4T6: ~/OSLAB/skill
GNU nano 8.7.1 memory1.c *
#include <stdio.h>
#include <stdlib.h>

int main() {
    int size = 5;

    int *buffer = (int *)malloc(size * sizeof(int));

    if (buffer == NULL) {
        printf("Memory allocation failed.\n");
        return 1;
    }

    for (int i = 0; i < size; i++) {
        buffer[i] = i + 1;
    }

    printf("Original buffer size: %d\n", size);

    size = 10;

    int *temp = (int *)realloc(buffer, size * sizeof(int));

    if (temp == NULL) {
        printf("Memory reallocation failed.\n");
        free(buffer);
        return 1;
    }

    buffer = temp;

    for (int i = 5; i < size; i++) {
        buffer[i] = i + 1;
    }

    printf("Resized buffer size: %d\n", size);

    printf("Buffer contents: ");

    for (int i = 0; i < size; i++) {
        printf("%d ", buffer[i]);
    }

    printf("\n");
}
```

^G Help ^O Write Out ^F Where Is ^K Cut ^T Execute ^C Location M-U Undo
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify ^_ Go To Line M-F Redo

```
monis@DESKTOP-C3GM4T6: ~/OSLAB/skill
GNU nano 8.7.1 memory1.c *

int *buffer = (int *)malloc(size * sizeof(int));

if (buffer == NULL) {
    printf("Memory allocation failed.\n");
    return 1;
}

for (int i = 0; i < size; i++) {
    buffer[i] = i + 1;
}

printf("Original buffer size: %d\n", size);

size = 10;

int *temp = (int *)realloc(buffer, size * sizeof(int));

if (temp == NULL) {
    printf("Memory reallocation failed.\n");
    free(buffer);
    return 1;
}

buffer = temp;

for (int i = 5; i < size; i++) {
    buffer[i] = i + 1;
}

printf("Resized buffer size: %d\n", size);

printf("Buffer contents: ");

for (int i = 0; i < size; i++) {
    printf("%d ", buffer[i]);
}

printf("\n");

free(buffer);

return 0;
}

Write to File: memory1.c
^G Help      M-D DOS Format  M-A Append     M-B Backup File ^T Browse
^C Cancel    M-M Mac Format  M-P Prepend    ^O Discard buffer
```

```
monis@DESKTOP-C3GM4T6: ~/OSLAB/skill
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$ nano memory1.c
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$ gcc memory1.c -o memory1
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$ ./memory1
Original buffer size: 5
Resized buffer size: 10
Buffer contents: 1 2 3 4 5 6 7 8 9 10
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$
```

Step 15: Prevent Buffer Overflow

monis@DESKTOP-C3GM4T6: ~/OSLAB/skill

```
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$ nano memory2.c
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$
```

monis@DESKTOP-C3GM4T6: ~/OSLAB/skill

```
GNU nano 8.7.1 memory2.c *
#include <stdio.h>
#include <stdlib.h>

int main() {
    int size = 5;

    int *buffer = malloc(size * sizeof(int));

    if (buffer == NULL) {
        printf("Memory allocation failed.\n");
        return 1;
    }

    int count = 0;

    for (int i = 1; i <= 7; i++) {
        if (count >= size) {
            printf("Buffer full. Resizing buffer...\n");

            int new_size = size * 2;

            int *temp = realloc(buffer, new_size * sizeof(int));

            if (temp == NULL) {
                printf("Memory reallocation failed.\n");
                free(buffer);
                return 1;
            }

            buffer = temp;
            size = new_size;
        }

        buffer[count] = i;
        count++;
    }

    printf("Final buffer size: %d\n", size);
    printf("Number of elements: %d\n", count);

    printf("Buffer contents: ");

    for (int i = 0; i < count; i++) {
        printf("%d ", buffer[i]);
    }
}
```

^G Help	^O Write Out	^F Where Is	^K Cut	^T Execute	^C Location	M-U Undo
^X Exit	^R Read File	^_\ Replace	^U Paste	^J Justify	^/_ Go To Line	M-E Redo

```
monis@DESKTOP-C3GM4T6: ~/OSLAB/skill
GNU nano 8.7.1 memory2.c *
    printf("Memory allocation failed.\n");
    return 1;
}

int count = 0;

for (int i = 1; i <= 7; i++) {

    if (count >= size) {
        printf("Buffer full. Resizing buffer...\n");

        int new_size = size * 2;

        int *temp = realloc(buffer, new_size * sizeof(int));

        if (temp == NULL) {
            printf("Memory reallocation failed.\n");
            free(buffer);
            return 1;
        }

        buffer = temp;
        size = new_size;
    }

    buffer[count] = i;
    count++;
}

printf("Final buffer size: %d\n", size);
printf("Number of elements: %d\n", count);

printf("Buffer contents: ");

for (int i = 0; i < count; i++) {
    printf("%d ", buffer[i]);
}

printf("\n");

free(buffer);

return 0;
}

Write to File: memory2.c
^G Help      M-D DOS Format  M-A Append      M-B Backup File ^T Browse
^C Cancel    M-M Mac Format  M-P Prepend     ^O Discard buffer
```

```
monis@DESKTOP-C3GM4T6: ~/OSLAB/skill
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$ nano memory2.c
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$ gcc memory2.c -o memory2
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$ ./memory2
Buffer full. Resizing buffer...
Final buffer size: 10
Number of elements: 7
Buffer contents: 1 2 3 4 5 6 7
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$
```

Step 16: Manage Linked Lists

Now demonstrate dynamic memory using a linked list.

monis@DESKTOP-C3GM4T6: ~/OSLAB/skill

```
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$ nano memory3.c
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$
```

```
monis@DESKTOP-C3GM4T6: ~/OSLAB/skill
GNU nano 8.7.1 memory3.c *
#include <stdio.h>
#include <stdlib.h>

struct Node {
    int data;
    struct Node *next;
};

int main() {

    struct Node *head = NULL;
    struct Node *second = NULL;
    struct Node *third = NULL;

    head = malloc(sizeof(struct Node));
    second = malloc(sizeof(struct Node));
    third = malloc(sizeof(struct Node));

    if (head == NULL || second == NULL || third == NULL) {
        printf("Memory allocation failed.\n");
        return 1;
    }

    head->data = 10;
    head->next = second;

    second->data = 20;
    second->next = third;

    third->data = 30;
    third->next = NULL;

    printf("Linked List: ");

    struct Node *current = head;

    while (current != NULL) {
        printf("%d ", current->data);
        current = current->next;
    }

    printf("\n");

    free(head);
    free(second);

^G Help      ^O Write Out ^F Where Is  ^K Cut       ^T Execute   ^C Location  M-U Undo
^X Exit      ^R Read File ^\ Replace   ^U Paste     ^J Justify   ^_ Go To Line M-E Redo
```

```
monis@DESKTOP-C3GM4T6: ~/OSLAB/skill
GNU nano 8.7.1 memory3.c *
};
struct Node *next;

int main() {

    struct Node *head = NULL;
    struct Node *second = NULL;
    struct Node *third = NULL;

    head = malloc(sizeof(struct Node));
    second = malloc(sizeof(struct Node));
    third = malloc(sizeof(struct Node));

    if (head == NULL || second == NULL || third == NULL) {
        printf("Memory allocation failed.\n");
        return 1;
    }

    head->data = 10;
    head->next = second;

    second->data = 20;
    second->next = third;

    third->data = 30;
    third->next = NULL;

    printf("Linked List: ");

    struct Node *current = head;

    while (current != NULL) {
        printf("%d ", current->data);
        current = current->next;
    }

    printf("\n");

    free(head);
    free(second);
    free(third);

    return 0;
}

Write to File: memory3.c
^G Help      M-D DOS Format  M-A Append     M-B Backup File ^T Browse
^C Cancel    M-M Mac Format  M-P Prepend    ^O Discard buffer
```

```
monis@DESKTOP-C3GM4T6: ~/OSLAB/skill
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$ nano memory3.c
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$ gcc memory3.c -o memory3
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$ ./memory3
Linked List: 10 20 30
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$
```

Step 17: Release Memory Correctly

The memory allocated using malloc() and realloc() must be released using free().

```
monis@DESKTOP-C3GM4T6: ~/OSLAB/skill
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$ nano memory4.c
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$
```

```
GNU nano 8.7.1 memory4.c *
#include <stdio.h>
#include <stdlib.h>

struct Node {
    int data;
    struct Node *next;
};

int main() {

    struct Node *head = NULL;
    struct Node *second = NULL;
    struct Node *third = NULL;

    /* Allocate memory for three nodes */
    head = malloc(sizeof(struct Node));
    second = malloc(sizeof(struct Node));
    third = malloc(sizeof(struct Node));

    /* Check memory allocation */
    if (head == NULL || second == NULL || third == NULL) {
        printf("Memory allocation failed.\n");

        free(head);
        free(second);
        free(third);

        return 1;
    }

    /* Store data in the nodes */
    head->data = 10;
    head->next = second;

    second->data = 20;
    second->next = third;

    third->data = 30;
    third->next = NULL;

    /* Display the linked list */
    printf("Linked List: ");

    struct Node *current = head;

^G Help      ^O Write Out ^F Where Is  ^K Cut       ^T Execute   ^C Location  M-U Undo
^X Exit      ^R Read File ^\ Replace   ^U Paste     ^J Justify   ^/ Go To Line M-E Redo
```

```
monis@DESKTOP-C3GM4T6: ~/OSLAB/skill
GNU nano 8.7.1 memory4.c *
third = malloc(sizeof(struct Node));

/* Check memory allocation */
if (head == NULL || second == NULL || third == NULL) {
    printf("Memory allocation failed.\n");

    free(head);
    free(second);
    free(third);

    return 1;
}

/* Store data in the nodes */
head->data = 10;
head->next = second;

second->data = 20;
second->next = third;

third->data = 30;
third->next = NULL;

/* Display the linked list */
printf("Linked List: ");

struct Node *current = head;

while (current != NULL) {
    printf("%d ", current->data);
    current = current->next;
}

printf("\n");

/* Release dynamically allocated memory */
free(head);
free(second);
free(third);

printf("Memory released successfully.\n");

return 0;
}

Write to File: memory4.c
^G Help      M-D DOS Format  M-A Append      M-B Backup File  ^T Browse
^C Cancel    M-M Mac Format  M-P Prepend     ^O Discard buffer
```

```
monis@DESKTOP-C3GM4T6: ~/OSLAB/skill
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$ nano memory4.c
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$ gcc memory4.c -o memory4
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$ ./memory4
Linked List: 10 20 30
Memory released successfully.
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$
```

Step 18: Verify with Valgrind

Step 19: Install Valgrind

In the same terminal, run:

sudo apt update

Wait for it to finish.

Then:

sudo apt install valgrind

If it asks:

Do you want to continue? [Y/n]

type:

Y

and press Enter.

```
monis@DESKTOP-C3GM4T6: ~/OSLAB/skill
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$ valgrind --version
Command 'valgrind' not found, but can be installed with:
sudo snap install valgrind # version 3.26.0, or
sudo apt install valgrind # version 1:3.26.0-0ubuntu1
See 'snap info valgrind' for additional versions.
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$ sudo apt update
[sudo: authenticate] Password:
Get:1 http://security.ubuntu.com/ubuntu resolute-security InRelease [137 kB]
Hit:2 http://archive.ubuntu.com/ubuntu resolute InRelease
Get:3 http://archive.ubuntu.com/ubuntu resolute-updates InRelease [137 kB]
Get:4 http://security.ubuntu.com/ubuntu resolute-security/main amd64 Packages [475 kB]
Get:5 http://archive.ubuntu.com/ubuntu resolute-backports InRelease [137 kB]
Get:6 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 Packages [611 kB]
Get:7 http://archive.ubuntu.com/ubuntu resolute-updates/main Translation-en [146 kB]
Get:8 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 Components [97.8 kB]
Get:9 http://archive.ubuntu.com/ubuntu resolute-updates/universe amd64 Packages [292 kB]
Get:10 http://archive.ubuntu.com/ubuntu resolute-updates/universe amd64 Components [188 kB]
Get:11 http://archive.ubuntu.com/ubuntu resolute-backports/universe amd64 Components [1056 B]
Get:12 http://security.ubuntu.com/ubuntu resolute-security/main Translation-en [112 kB]
Get:13 http://security.ubuntu.com/ubuntu resolute-security/main amd64 Components [46.6 kB]
Get:14 http://security.ubuntu.com/ubuntu resolute-security/universe amd64 Components [43.4 kB]
Fetched 2424 kB in 19s (126 kB/s)
116 packages can be upgraded. Run 'apt list --upgradable' to see them.
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$ sudo apt install valgrind
Upgrading:
  libc-bin libc-dev-bin libc-gconv-modules-extra libc6 libc6-dev locales

Installing:
  valgrind

Installing dependencies:
  gdb libdebuginfod-common libipt2
  libbabeltrace1 libdebuginfod1t64 libsource-highlight-common
  libc6-dbg libdw1t64 libsource-highlight4t64

Suggested packages:
  gdb-doc gdbserver valgrind-mpi kcachegrind

Summary:
  Upgrading: 6, Installing: 10, Removing: 0, Not Upgrading: 110
  Download size: 27.1 MB / 37.8 MB
  Space needed: 111 MB / 1024 GB available

Continue? [Y/n] y
Get:1 http://archive.ubuntu.com/ubuntu resolute/main amd64 libdebuginfod-common all 0.194-4 [12.3 kB]
Get:2 http://archive.ubuntu.com/ubuntu resolute/main amd64 libdw1t64 amd64 0.194-4 [285 kB]
Get:3 http://archive.ubuntu.com/ubuntu resolute/main amd64 libbabeltrace1 amd64 1.5.11-5build1 [165 kB]
Get:4 http://archive.ubuntu.com/ubuntu resolute/main amd64 libdebuginfod1t64 amd64 0.194-4 [20.7 kB]
Get:5 http://archive.ubuntu.com/ubuntu resolute/main amd64 libipt2 amd64 2.1-2-2 [40.2 kB]
```

```
Continue? [Y/n] y
Get:1 http://archive.ubuntu.com/ubuntu resolute/main amd64 libdebuginfod-common all 0.194-4 [12.3 kB]
Get:2 http://archive.ubuntu.com/ubuntu resolute/main amd64 libdw1t64 amd64 0.194-4 [285 kB]
Get:3 http://archive.ubuntu.com/ubuntu resolute/main amd64 libbabeltrace1 amd64 1.5.11-5build1 [165 kB]
Get:4 http://archive.ubuntu.com/ubuntu resolute/main amd64 libdebuginfod1t64 amd64 0.194-4 [20.7 kB]
Get:5 http://archive.ubuntu.com/ubuntu resolute/main amd64 libipt2 amd64 2.1.2-3 [49.2 kB]
Get:6 http://archive.ubuntu.com/ubuntu resolute/main amd64 libsource-highlight-common all 3.1.9-4.3build2 [64.2 kB]
Get:7 http://archive.ubuntu.com/ubuntu resolute/main amd64 libsource-highlight4t64 amd64 3.1.9-4.3build2 [273 kB]
Get:8 http://archive.ubuntu.com/ubuntu resolute/main amd64 gdb amd64 17.1-2ubuntu1 [4173 kB]
Get:9 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 libc6-dbg amd64 2.43-2ubuntu2.4 [6406 kB]
Get:10 http://archive.ubuntu.com/ubuntu resolute/main amd64 valgrind amd64 1:3.26.0-0ubuntu1 [15.7 MB]
Fetched 27.1 MB in 10s (2805 kB/s)
Preconfiguring packages ...
(Reading database ... 42276 files and directories currently installed.)
Preparing to unpack .../libc-dev-bin_2.43-2ubuntu2.4_amd64.deb ...
Unpacking libc-dev-bin (2.43-2ubuntu2.4) over (2.43-2ubuntu2.3) ...
Preparing to unpack .../libc6-dev_2.43-2ubuntu2.4_amd64.deb ...
Unpacking libc6-dev:amd64 (2.43-2ubuntu2.4) over (2.43-2ubuntu2.3) ...
Preparing to unpack .../libc-gconv-modules-extra_2.43-2ubuntu2.4_amd64.deb ...
Unpacking libc-gconv-modules-extra:amd64 (2.43-2ubuntu2.4) over (2.43-2ubuntu2.3) ...
Setting up libc-gconv-modules-extra:amd64 (2.43-2ubuntu2.4) ...
(Reading database ... 42276 files and directories currently installed.)
Preparing to unpack .../libc6_2.43-2ubuntu2.4_amd64.deb ...
Unpacking libc6:amd64 (2.43-2ubuntu2.4) over (2.43-2ubuntu2.3) ...
Setting up libc6:amd64 (2.43-2ubuntu2.4) ...
(Reading database ... 42276 files and directories currently installed.)
Preparing to unpack .../libc-bin_2.43-2ubuntu2.4_amd64.deb ...
Unpacking libc-bin (2.43-2ubuntu2.4) over (2.43-2ubuntu2.3) ...
Setting up libc-bin (2.43-2ubuntu2.4) ...
ldconfig: /usr/lib/wsl/lib/libcuda.so.1 is not a symbolic link

Selecting previously unselected package libdebuginfod-common.
(Reading database ... 42276 files and directories currently installed.)
Preparing to unpack .../00-libdebuginfod-common_0.194-4_all.deb ...
Unpacking libdebuginfod-common (0.194-4) ...
Preparing to unpack .../01-locales_2.43-2ubuntu2.4_all.deb ...
Unpacking locales (2.43-2ubuntu2.4) over (2.43-2ubuntu2.3) ...
Selecting previously unselected package libdw1t64:amd64.
Preparing to unpack .../02-libdw1t64_0.194-4_amd64.deb ...
Unpacking libdw1t64:amd64 (0.194-4) ...
Selecting previously unselected package libbabeltrace1:amd64.
Preparing to unpack .../03-libbabeltrace1_1.5.11-5build1_amd64.deb ...
Unpacking libbabeltrace1:amd64 (1.5.11-5build1) ...
Selecting previously unselected package libdebuginfod1t64:amd64.
Preparing to unpack .../04-libdebuginfod1t64_0.194-4_amd64.deb ...
Unpacking libdebuginfod1t64:amd64 (0.194-4) ...
```

```
monis@DESKTOP-C3GM4T6: ~/OSLAB/skill
Unpacking libdebuginfod1t64:amd64 (0.194-4) ...
Selecting previously unselected package libipt2.
Preparing to unpack .../05-libipt2_2.1.2-3_amd64.deb ...
Unpacking libipt2 (2.1.2-3) ...
Selecting previously unselected package libsource-highlight-common.
Preparing to unpack .../06-libsource-highlight-common_3.1.9-4.3build2_all.deb ...
Unpacking libsource-highlight-common (3.1.9-4.3build2) ...
Selecting previously unselected package libsource-highlight4t64:amd64.
Preparing to unpack .../07-libsource-highlight4t64_3.1.9-4.3build2_amd64.deb ...
Unpacking libsource-highlight4t64:amd64 (3.1.9-4.3build2) ...
Selecting previously unselected package gdb.
Preparing to unpack .../08-gdb_17.1-2ubuntu1_amd64.deb ...
Unpacking gdb (17.1-2ubuntu1) ...
Selecting previously unselected package libc6-dbg:amd64.
Preparing to unpack .../09-libc6-dbg_2.43-2ubuntu2.4_amd64.deb ...
Unpacking libc6-dbg:amd64 (2.43-2ubuntu2.4) ...
Selecting previously unselected package valgrind.
Preparing to unpack .../10-valgrind_1%3a3.26.0-0ubuntu1_amd64.deb ...
Unpacking valgrind (1:3.26.0-0ubuntu1) ...
Setting up libdebuginfod-common (0.194-4) ...
Setting up locales (2.43-2ubuntu2.4) ...
Generating locales (this might take a while)...
Generation complete.
Setting up libsource-highlight-common (3.1.9-4.3build2) ...
Setting up libc6-dbg:amd64 (2.43-2ubuntu2.4) ...
Setting up libdw1t64:amd64 (0.194-4) ...
Setting up libipt2 (2.1.2-3) ...
Setting up libbabeltrace1:amd64 (1.5.11-5build1) ...
Setting up libc-dev-bin (2.43-2ubuntu2.4) ...
Setting up valgrind (1:3.26.0-0ubuntu1) ...
Setting up libdebuginfod1t64:amd64 (0.194-4) ...
Setting up libsource-highlight4t64:amd64 (3.1.9-4.3build2) ...
Setting up gdb (17.1-2ubuntu1) ...
Setting up libc6-dev:amd64 (2.43-2ubuntu2.4) ...
Processing triggers for systemd (259.5-0ubuntu3.4) ...
Processing triggers for man-db (2.13.1-1build1) ...
Processing triggers for libc-bin (2.43-2ubuntu2.4) ...
ldconfig: /usr/lib/wsl/lib/libcuda.so.1 is not a symbolic link
```

Step 20: Check Valgrind

After installation, run:

valgrind --version

You should get something similar to:

valgrind-3.xx.x

```
monis@DESKTOP-C3GM4T6: ~/OSLAB/skill
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$ valgrind --version
valgrind-3.26.0
```

Step 21: Compile your program for Valgrind

Your file is memory4.c, so run:

gcc -g memory4.c -o memory4

Then run:

valgrind --leak-check=full ./memory4

What you want to see

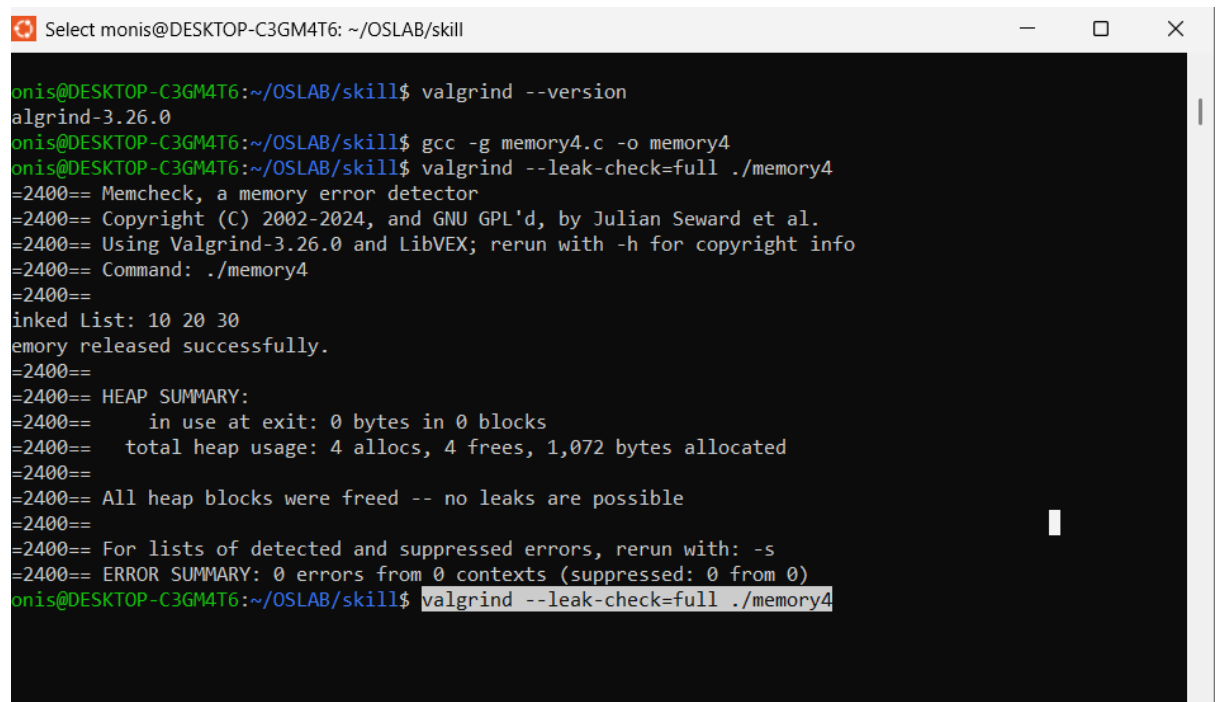
At the bottom, you should see something similar to:

HEAP SUMMARY:

in use at exit: 0 bytes in 0 blocks

total heap usage: 3 allocs, 3 frees, ...

All heap blocks were freed -- no leaks are possible
The exact numbers may differ.



```
Select monis@DESKTOP-C3GM4T6: ~/OSLAB/skill

monis@DESKTOP-C3GM4T6:~/OSLAB/skill$ valgrind --version
valgrind-3.26.0
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$ gcc -g memory4.c -o memory4
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$ valgrind --leak-check=full ./memory4
==2400== Memcheck, a memory error detector
==2400== Copyright (C) 2002-2024, and GNU GPL'd, by Julian Seward et al.
==2400== Using Valgrind-3.26.0 and LibVEX; rerun with -h for copyright info
==2400== Command: ./memory4
==2400==
==2400== linked List: 10 20 30
==2400== memory released successfully.
==2400==
==2400== HEAP SUMMARY:
==2400==      in use at exit: 0 bytes in 0 blocks
==2400==    total heap usage: 4 allocs, 4 frees, 1,072 bytes allocated
==2400==
==2400== All heap blocks were freed -- no leaks are possible
==2400==
==2400== For lists of detected and suppressed errors, rerun with: -s
==2400== ERROR SUMMARY: 0 errors from 0 contexts (suppressed: 0 from 0)
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$ valgrind --leak-check=full ./memory4
```